

WHATSNEXT VISION MOTORS

Shaping the Future of Mobility with Innovation and Excellence

Salesforce CRM Implementation

Vehicle Ordering • Dealer Assignment • Stock Validation • Test Drive Automation

Project: WhatsNext Vision Motors CRM

Platform: Salesforce Developer Org

Core Technologies: Custom Objects • Flow • Apex • Batch Apex • Scheduled Apex

Primary Goal: Automate vehicle ordering and improve customer experience

Project Model / Implementation Guide

1. Project Overview

WhatsNext Vision Motors is implementing Salesforce CRM to streamline vehicle ordering, dealer operations, stock management, and test-drive communication. The solution uses Salesforce configuration plus Apex automation.

Orders are validated against available vehicle stock, confirmed orders reduce stock, pending orders can be processed automatically when stock becomes available, and scheduled reminders are sent for upcoming test drives.

Key business capabilities:

- Maintain Vehicle, Vehicle Dealer, Customer, Vehicle Order and Vehicle Test Drive information.
- Provide a dedicated Salesforce Lightning application named WhatNext Vision Motors.
- Prevent orders when the selected vehicle has no available stock.
- Reduce vehicle stock when an order is confirmed.
- Automatically process pending orders when new stock becomes available.
- Send an email reminder before a scheduled test drive.
- Run batch order processing on a recurring schedule.

High-level automation flow: Customer → Vehicle Order → Stock Validation → Confirmation → Stock Update

2. Creating the Salesforce Developer Account

Create a Salesforce Developer Org before starting the configuration.

Signup URL: <https://developer.salesforce.com/signup>

Enter the following details:

- First Name / Last Name: Enter your name.
- Email: Use an accessible email address.
- Role: Developer.
- Company: College name.
- Country: India.
- Postal Code: Your PIN code.
- Username: Use a unique Salesforce username.

Click Sign Me Up. Verify the Salesforce email, set the password/security details, and log in to the Developer Org.

3. Creating the Vehicle Dealer Custom Object

Navigation: Setup → Object Manager → Create → Custom Object

Setting	Value
Label	Vehicle Dealer
Plural Label	Vehicle Dealers
Record Name Label	Dealer Name
Data Type	Text
Additional Options	Allow Reports; Allow Search

4. Creating the Lightning App

Create a dedicated Lightning App so the project objects and reports can be accessed from one navigation menu.

Navigation: Setup → Quick Find → App Manager → New Lightning App

App Name: **WhatNext Vision Motors**

Description: Salesforce CRM application for vehicle sales and customer operations

Wizard steps:

- App Details → enter the application information.
- App Options → keep defaults → Next.
- Utility Items → keep defaults → Next.
- Navigation Items → add Vehicle, Vehicle Dealer, Customer, Order, Test Drive, Service Request, Reports and Dashboard.
- Select the custom Vehicle Dealer object created in the previous step.
- Add the System Administrator profile under User Profiles as required.
- Finish the app configuration and save.

5. Record-Triggered Flow — Test Drive Reminder

Create a record-triggered flow that sends a customer an email reminder one day before a scheduled test drive.

Trigger: Vehicle Test Drive; record is created or updated; Status__c = Scheduled.

Scheduled Path: Time Source = Test_Drive_Date__c; Offset = 1 Day Before.

Get Customer Information: Object = Vehicle_Customer__c; Id = {!\$Record.Customer__c}.

Send Email: Subject = “Reminder: Your Test Drive is Tomorrow!”; recipient = customer's Email__c.

Save the flow as **Test Drive Reminder**, activate it, and test it with a future scheduled test drive.

6. Apex Trigger Handler — Stock Validation

The trigger handler separates business logic from the trigger. Before insert/update, it validates stock. After insert/update, it updates stock when an order is confirmed.

Class: VehicleOrderTriggerHandler

Core logic:

- Collect Vehicle__c IDs from incoming orders.
- Query Vehicle__c records and check Stock_Quantity__c.
- If stock is null or less than or equal to zero, add an error: "This vehicle is out of stock. Order cannot be placed."
- For confirmed orders, reduce Stock_Quantity__c by 1.
- Update the affected Vehicle records.

7. Vehicle Order Trigger

Create the trigger on Vehicle_Order__c. It calls the handler for before/after insert and update events.

Events: before insert, before update, after insert, after update.

Handler: VehicleOrderTriggerHandler.handleTrigger(...)

Why this design?

- Trigger: handles Salesforce events only.
- Handler: contains reusable business logic.
- Before event: stops an invalid/out-of-stock order with addError().
- After event: reduces vehicle stock after a confirmed order.

8. Batch Apex — Process Pending Orders

A pending order can remain unconfirmed when the vehicle is unavailable. The batch job periodically checks pending orders and confirms them when stock becomes available.

Class: VehicleOrderBatch

Processing sequence:

- Query Vehicle_Order__c records where Status__c = 'Pending'.
- Collect the related vehicle IDs.
- Query current vehicle stock.
- For vehicles with stock greater than zero, set the order status to Confirmed.
- Decrease vehicle stock by 1.
- Update the orders and vehicles.

A batch size such as 50 can be supplied when the batch is started.

9. Scheduled Apex — Run the Batch Automatically

Create a schedulable class that launches VehicleOrderBatch.

Class: VehicleOrderBatchScheduler

Schedule the job from Execute Anonymous or an equivalent Apex execution interface.

Daily schedule: 0 0 0 * * ? — every day at 12:00 AM (midnight).

Important: 0 0 12 * * ? runs at 12:00 PM (noon).

Verify the schedule in Setup → Scheduled Jobs and check the job name, next execution time, and status.

10. Testing Checklist

Test Case	Expected Result
Create Vehicle with stock > 0	Vehicle record saves successfully.
Create order for in-stock vehicle	Order can proceed according to configured status/automation.
Create order when stock = 0	Trigger prevents the order and displays the out-of-stock message.
Confirm an order	Vehicle stock decreases by 1.
Create Pending order	Order remains Pending while stock is unavailable.
Add new vehicle stock	Scheduled batch can find the pending order.
Run VehicleOrderBatch	Eligible pending order becomes Confirmed and stock decreases.
Create scheduled test drive	Flow schedules the reminder one day before the test drive.
Reach reminder time	Customer receives the configured email reminder.
Check Scheduled Jobs	Daily Vehicle Order Processing appears with its next run.

Recommended validation sequence:

- Verify all custom objects and required fields exist with the exact API names used by Apex.
- Activate the Flow and test the email path.
- Test the trigger with both stock available and stock unavailable.
- Run the batch manually once, then schedule it and verify the Scheduled Jobs entry.

11. Solution Architecture

Layer	Salesforce Component	Responsibility
Data	Vehicle / Dealer / Customer / Order / Test Drive	Stores CRM records.
UI	Lightning App	Central navigation for project users.
Automation	Record-Triggered Flow	Schedules and sends test-drive reminders.
Validation	Apex Trigger + Handler	Validates stock and updates stock.
Processing	Batch Apex	Processes pending orders in bulk.
Scheduling	Schedulable Apex	Runs the batch automatically.

Business Outcome

The completed design creates a connected CRM workflow: customers and vehicles are represented as Salesforce records; order events invoke Apex validation; confirmed orders adjust inventory; pending orders can be processed later through Batch Apex; and scheduled Flow automation keeps customers informed about test drives.

Implementation Note

The Apex examples use standard Salesforce custom-field API naming conventions such as Vehicle__c, Status__c and Stock_Quantity__c. Confirm the exact API names in the Salesforce Developer Org before saving code.

12. Completion Summary

- Salesforce Developer Org setup
- Vehicle Dealer custom object
- WhatNext Vision Motors Lightning App
- Test Drive Reminder record-triggered Flow
- Vehicle Order Apex Trigger + Handler
- Vehicle Order Batch Apex
- Scheduled Apex for daily processing
- Testing and verification checklist

End of Project Model Documentation